TIANCAN XIA

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Education

Southeast University (SEU) **Nanjing, China**
Master of Science in Information and Communication Engineering Sep 2024 - Jun 2027
GPA: 4.0/4.0 | **Average Score**: 94.77/100 | **Ranking**: 1/142

Southeast University (SEU) **Nanjing, China**
Bachelor of Engineering in Information Engineering Sep 2020 - Jun 2024
GPA: 3.98/4.0 | **Average Score**: 92.49/100 | **Ranking**: 4/258

Research Interests

Core Focus: Generative models, probabilistic graphical models
Methodologies: Diffusion models, Transformer architectures, variational methods
Problem Domains: Inverse problems, structured Bayesian inference

Publications

[1] **Tiancan Xia**, Jian Zheng, Xiaosi Tan, and Chuan Zhang, "Message Passing-Based Diffusion Posterior Sampling for Generalized Linear Inverse Problems," Manuscript completed.

[2] **Tiancan Xia**, Jian Zheng, Xiaosi Tan, Yongming Huang, Xiaohu You, and Chuan Zhang, "Variational Bayesian Message Passing Receiver for Uplink ISAC Systems," *IEEE Transactions on Communications*, to appear.

[3] **Tiancan Xia**, Xiaosi Tan, Yongming Huang, Xiaohu You, and Chuan Zhang, "Jointly-Layered Iterative Detection and Decoding for Massive MIMO: An EP-SU Based Framework," *IEEE Wireless Communications Letters*, vol. 15, pp. 1440-1444, Jan. 2026.

[4] **Tiancan Xia**, Xiaosi Tan, Wenyue Zhou, Yongming Huang, Xiaohu You, and Chuan Zhang, "EP-based Distributed Message Passing Detection for Cell-Free Massive MIMO," *IEEE Wireless Communications Letters*, vol. 15, pp. 495-499, Jan. 2026.

[5] **Tiancan Xia**, Xiaohua Xie, Xiaosi Tan, Yongming Huang, Xiaohu You, and Chuan Zhang, "An Efficient Iterative Detection and Decoding Receiver for Polar-Coded Massive MIMO," *Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, Oct. 2023, pp. 1505-1509.

[6] Yu Tian, Jian Zheng, Zongyao Li, **Tiancan Xia**, Yifei Shen, Wenyue Zhou, Xiaosi Tan, Jiamin Li, Yongming Huang, Chuan Zhang, and Xiaohu You, "Turbo Product Code-Aided Spatiotemporal 2-D Coded MIMO for URLLC," *IEEE Communications Magazine*, to appear.

[7] Jian Zheng, Yi Sun, **Tiancan Xia**, Wenyue Zhou, Xiaosi Tan, Yongming Huang, and Chuan Zhang, "AttnMPNet: An Attention-Based Message Passing Network for MIMO Detection," *IEEE Wireless Communications Letters*, vol. 15, pp. 1861-1865, Feb. 2026.

[8] Houren Ji, Xiaohua Xie, **Tiancan Xia**, Xiaosi Tan, Yongming Huang, Xiaohu You, and Chuan Zhang, "Low-Complexity Double EP Approximation for Parallel Detection-Decoding in MIMO Systems," *IEEE Wireless Communications Letters*, vol. 14, no. 11, pp. 3395-3399, Nov. 2025.

[9] Xiaosi Tan, Xiaohua Xie, Houren Ji, **Tiancan Xia**, Yongming Huang, Xiaohu You, and Chuan Zhang, "A Soft Iterative Receiver with Simplified EP Detection for Coded MIMO Systems," *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 33, no. 7, pp. 1994-1998, July 2025.

Research Experience

LEADS, Southeast University **Nanjing, China**
Research Assistant — Lab of Efficient Architectures for Digital-communication and Signal-processing
Supervisors: Prof. Xiaosi Tan, Prof. Chuan Zhang

Diffusion Posterior Sampling for Generalized Linear Inverse Problems 2025 – present
Proposed a diffusion posterior sampling framework integrating message passing with pre-trained diffusion models

- Reformulated posterior score estimation as an MMSE inference problem via the generalized Tweedie formula, eliminating the need to backpropagate through the forward operator for likelihood score approximation
- Integrated diffusion timestep-dependent states as prior inputs to message passing iterations, enabling the inference procedure to leverage evolving generative priors
- Developed a modular framework where the message passing algorithm, score approximation technique, and diffusion sampler can each be independently configured and upgraded
- Demonstrated competitive zero-shot reconstruction on compressive image recovery benchmarks

Attention and Transformer Architectures for High-Dimensional Bayesian Inference 2025 – present

Proposed attention-augmented message passing and Transformer architectures for iterative posterior inference

- Designed the feature representation scheme that integrates Gaussian approximate message passing with self-attention for iterative posterior refinement, validated in massive MIMO detection scenarios
- Extended the framework to joint parameter estimation and signal recovery, incorporating continuous parameter inference alongside discrete symbol recovery
- Formulated a dual-stream Transformer architecture on bipartite factor graphs for generalized Bayesian inference

Variational Inference for Estimation and Detection in Linear Systems 2023 – 2026

Proposed variational message passing methods leveraging system model structure

- Developed a mean-field variational inference framework for joint parameter estimation and signal detection, validated in integrated sensing and communication scenarios
- Extended the mean-field formulation to Bethe variational approximation, deriving closed-form update equations via stationary point analysis of the Bethe free energy
- Incorporated noise variance learning through free energy minimization, with convergence and estimation accuracy characterized through state evolution analysis
- Developed low-complexity expectation propagation-based message passing algorithms for distributed systems, validated in cell-free massive MIMO networks

Factor Graph-based Iterative Inference for Joint Detection and Decoding 2022 – 2025

Proposed expectation propagation-based iterative detection and decoding algorithms on joint factor graph models

- Developed an approximate expectation propagation-based iterative detection and decoding algorithm with a critical-set-based belief propagation decoder for polar-coded massive MIMO systems
- Designed a jointly-layered iterative architecture enabling mini-batch message exchange between detection and decoding modules on factor graphs for LDPC-coded massive MIMO systems
- Demonstrated improved error rate performance across varying scenarios compared to existing iterative methods

Academic Activities

Peer Reviewer:
IEEE Transactions on Communications (2 manuscripts), IEEE Open Journal of the Communications Society,
IEEE International Conference on Communications (ICC), IEEE Workshop on Signal Processing Systems (SiPS)

Conference Presentation:
Poster Presentation, 57th Asilomar Conference on Signals, Systems, and Computers, Pacific Grove, California, USA Oct. 2023

Honors & Awards

- **2025 China National Scholarship** (**Top 0.2%**, the highest honor for postgraduates in China)
- 2025 Merit Student Award (Graduate Level)
- 2024 First-Class Academic Scholarship
- **2024 Jiangsu Provincial Outstanding Undergraduate Thesis Award**
- **2023 China National Scholarship** (**Top 0.2%**, the highest honor for undergraduates in China)
- 2022&2023 Merit Student for two consecutive years
- 2022&2023 Outstanding Academic Performance Award for two consecutive years
- 2022 Jiangsu Province College Students Electronic Design Competition (Provincial **1st Prize**)
- 2021 China Undergraduate Mathematics Competition (Provincial **1st Prize**, National **2nd Prize**)

Selected Courses

Postgraduate:
Stochastic Process (100/100), Matrix Theory in Engineering (95/100), Advanced Digital Signal Processing (97/100)
Undergraduate:
Artificial Intelligence and Deep Learning (98/100), Statistical Signal Processing (97/100), Linear Algebra (100/100),
Probability Statistics & Stochastic Processes (98/100), Mathematical Analysis I/II (95/100), Signals and Systems (96/100)

Skills

Programming & Tools: Python, PyTorch, MATLAB, C/C++, LaTeX, Git, Verilog
Languages: Mandarin (Native), English (IELTS 8.0)